Cell Biology Lecture: Cells, Membranes, and Energy

Introduction

Cell biology is the study of the structure and function of cells. All living organisms are made of one or more cells, and all cells arise from existing cells. This is known as cell theory. Cells are the basic unit of life because they carry out the chemical reactions that sustain life.

There are two major categories of cells. Prokaryotic cells, such as bacteria, do not have a membrane-bound nucleus. Their genetic material is located in a region called the nucleoid. Eukaryotic cells, such as plant and animal cells, have a membrane-bound nucleus and specialized compartments called organelles. Organelles allow different chemical reactions to occur in separate environments.

The Plasma Membrane

The plasma membrane is a selectively permeable barrier that surrounds every cell. It is made mainly of a phospholipid bilayer. Each phospholipid has a hydrophilic phosphate head and two hydrophobic fatty acid tails. In water, the tails face inward and the heads face outward, forming a stable bilayer.

Membrane proteins perform many functions. Channel proteins allow specific ions to pass through the membrane. Carrier proteins change shape to move substances across the membrane. Receptor proteins bind signaling molecules and trigger responses inside the cell. The fluid mosaic model describes the membrane as a flexible mosaic of phospholipids, proteins, cholesterol, and carbohydrates.

Passive transport moves substances from an area of higher concentration to an area of lower concentration without using ATP. Diffusion and osmosis are examples of passive transport. Active transport moves substances against their concentration gradient and requires energy, usually from ATP. The sodium-potassium pump is an example of active transport: it moves three sodium ions out of the cell and two potassium ions into the cell for each ATP that is used.

Organelles and Their Functions

The nucleus contains most of the cell’s DNA and controls gene expression. It is surrounded by a double membrane called the nuclear envelope. The nucleolus, located inside the nucleus, is the site of ribosome assembly.

Ribosomes are the molecular machines that translate messenger RNA into proteins. They can be free in the cytoplasm or attached to the rough endoplasmic reticulum. The rough endoplasmic reticulum is a network of membranes that helps fold and modify proteins. The smooth endoplasmic reticulum makes lipids and helps detoxify harmful substances.

The Golgi apparatus modifies, sorts, and packages proteins and lipids for transport. It is often described as the cell’s post office. Lysosomes contain digestive enzymes that break down worn-out organelles, food particles, and invading bacteria. Vacuoles store water, nutrients, and waste; plant cells usually have a large central vacuole that helps maintain turgor pressure.

Mitochondria are the main site of aerobic respiration. They have an outer membrane and a highly folded inner membrane called the cristae. The folds increase the surface area available for the reactions that produce ATP. Mitochondria contain their own DNA and ribosomes, which supports the endosymbiotic theory: mitochondria may have evolved from free-living bacteria that were engulfed by an ancestral eukaryotic cell.

Chloroplasts are found in plant cells and some algae. They contain chlorophyll and carry out photosynthesis, converting light energy into chemical energy stored in glucose. Both mitochondria and chloroplasts have double membranes and their own DNA.

Energy and Metabolism

Cells need energy to maintain order, move substances, build molecules, and reproduce. ATP, or adenosine triphosphate, is the universal energy currency of the cell. ATP stores energy in its phosphate bonds. When the terminal phosphate bond is broken, ATP becomes ADP and inorganic phosphate, releasing energy that can be used for cellular work.

Cellular respiration is the process that converts glucose and oxygen into ATP, carbon dioxide, and water. It has three main stages: glycolysis, the Krebs cycle, and the electron transport chain. Glycolysis takes place in the cytoplasm and breaks one glucose molecule into two pyruvate molecules, producing a small amount of ATP and NADH.

The Krebs cycle takes place in the mitochondrial matrix. It completes the oxidation of pyruvate and produces NADH, FADH2, and a small amount of ATP. The electron transport chain is located in the inner mitochondrial membrane. Electrons from NADH and FADH2 pass through a series of protein complexes, pumping protons into the intermembrane space. The protons then flow back through ATP synthase, which produces most of the cell’s ATP. Oxygen is the final electron acceptor in this process.

Fermentation occurs when oxygen is not available. It allows glycolysis to continue by regenerating NAD+. Lactic acid fermentation produces lactate, while alcoholic fermentation produces ethanol and carbon dioxide. Fermentation produces much less ATP than aerobic respiration.

Photosynthesis is the reverse of cellular respiration in terms of energy flow. It uses light energy, carbon dioxide, and water to produce glucose and oxygen. The light-dependent reactions occur in the thylakoid membranes, and the Calvin cycle occurs in the stroma of the chloroplast.

Cell Division and Study Guide

Cells reproduce by dividing. In mitosis, a eukaryotic cell divides to produce two genetically identical daughter cells. Mitosis is used for growth, repair, and asexual reproduction. The cell cycle includes interphase, mitosis, and cytokinesis. Interphase is divided into G1, S, and G2 phases. DNA is replicated during the S phase.

Meiosis produces gametes with half the normal number of chromosomes. It involves two rounds of division and creates genetic variation through crossing over and independent assortment. Fertilization restores the diploid chromosome number.

Key facts to remember:
1. The plasma membrane is a phospholipid bilayer that controls what enters and leaves the cell.
2. Mitochondria produce most of the cell’s ATP through aerobic respiration.
3. Ribosomes translate mRNA into proteins.
4. The nucleus contains DNA and controls gene expression.
5. ATP is the universal energy currency of the cell.
6. Enzymes are biological catalysts that lower activation energy.
7. Prokaryotic cells lack a membrane-bound nucleus; eukaryotic cells have one.
8. Active transport requires ATP, while passive transport does not.
9. The electron transport chain produces the largest amount of ATP in aerobic respiration.
10. Photosynthesis converts light energy into chemical energy stored in glucose.

Review questions:
- What is the main function of the plasma membrane?
- Which organelle produces most of the ATP in a eukaryotic cell?
- What are the inputs and outputs of aerobic respiration?
- How do prokaryotic and eukaryotic cells differ?
- Where does translation take place?